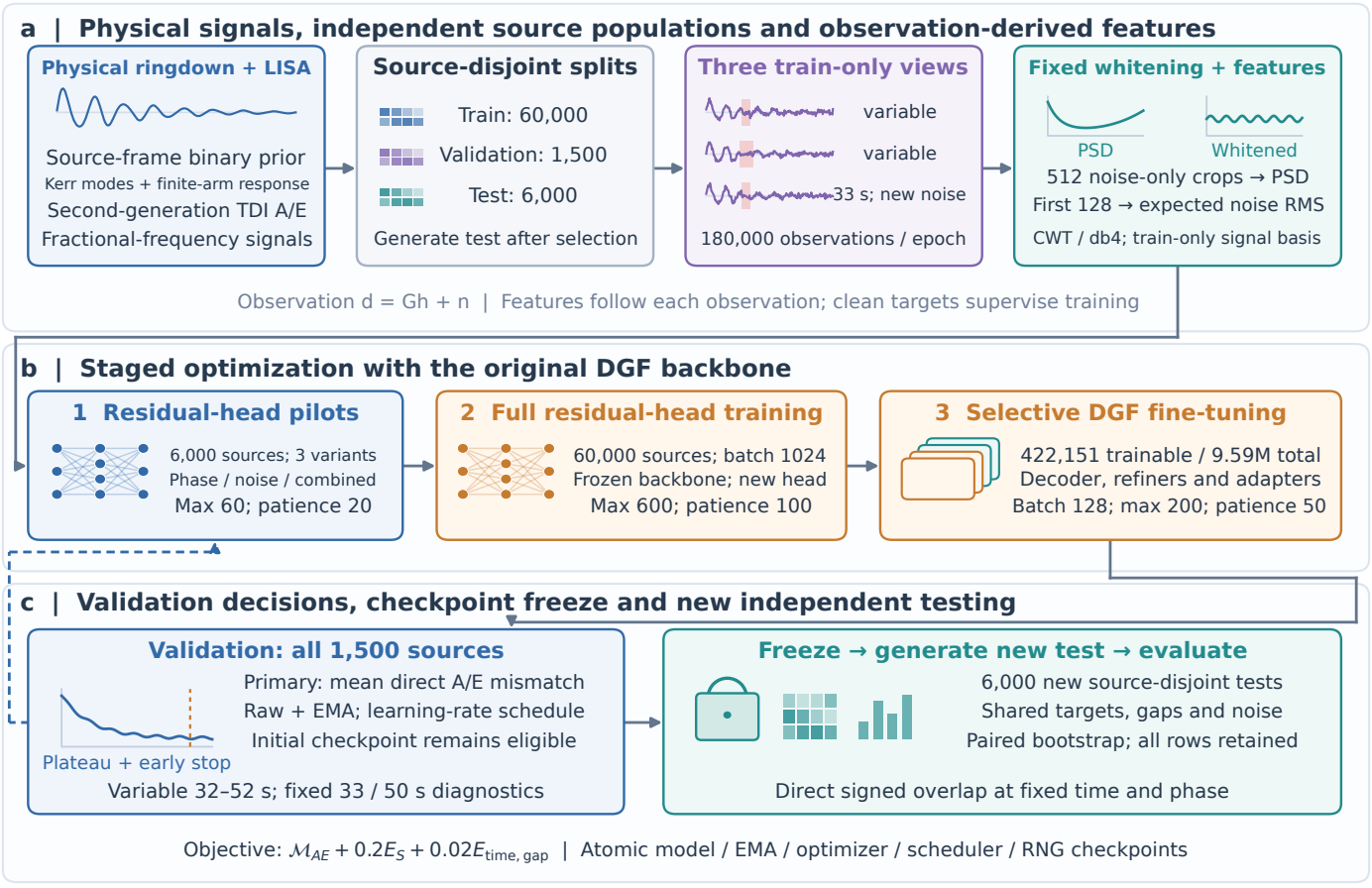
a | Physical signals, independent source populations and observation-derived features
Physical ringdown + LISA
Source-frame binary prior
Kerr modes + finite-arm response
Second-generation TDI A/E
Fractional-frequency signals
Source-disjoint splits
Train: 60,000
Validation: 1,500
Test: 6,000
Generate test after selection
Three train-only views
variable
variable
33 s; new noise
180,000 observations / epoch
Fixed whitening + features
PSD
Whitened
512 noise-only crops → PSD
First 128 → expected noise RMS
CWT / db4; train-only signal basis
Observation d = Gh + n | Features follow each observation; clean targets supervise training
b | Staged optimization with the original DGF backbone
1 Residual-head pilots
6,000 sources; 3 variants
Phase / noise / combined
Max 60; patience 20
2 Full residual-head training
60,000 sources; batch 1024
Frozen backbone; new head
Max 600; patience 100
3 Selective DGF fine-tuning
422,151 trainable / 9.59M total
Decoder, refiners and adapters
Batch 128; max 200; patience 50
c | Validation decisions, checkpoint freeze and new independent testing
Validation: all 1,500 sources
Primary: mean direct A/E mismatch
Raw + EMA; learning-rate schedule
Initial checkpoint remains eligible
Plateau + early stop
Variable 32–52 s; fixed 33 / 50 s diagnostics
Freeze → generate new test → evaluate
6,000 new source-disjoint tests
Shared targets, gaps and noise
Paired bootstrap; all rows retained
Direct signed overlap at fixed time and phase
Objective: $\mathcal{M}_{AE} + 0.2E_S + 0.02E_{\text{time, gap}}$ | Atomic model / EMA / optimizer / scheduler / RNG checkpoints